Suggested Comments/Key Message: The tool we will be discussing that is used to complete systemic analysis was developed in the above document by FHWA in 2013. The first document (on left on slide) walks you through the step by step process; describing the objectives, data needs and process for completing each element, step and task. In addition to the original guide, FHWA has also published some supplemental case studies that demonstrate how real agencies went about conducting systemic analysis in the document on the right. The two case studies described in the document include a pedestrian and bike focused systemic analysis and a systemic analysis completed with limited data.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf and https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/fhwasa17002.pdf

Suggested Comments/Key Message: As part of the document on the previous slide, FHWA developed a process for the systemic approach to safety planning (and related tool) to provide guidance to agencies so they can incorporate systemic improvements into their safety programs to increase the potential to reduce severe crashes. This tool follows a framework that is made up of the following three elements:

- Element 1: The Systemic Safety Project Selection Process. A step-by-step process to conduct systemic safety analysis and planning
- Element 2: Balancing Traditional and Systemic Safety Investments . This is a framework which includes analytical techniques for determining a reasonable distribution between the implementation of spot safety improvements and systemic safety improvements
- Element 3: Performing Systemic Program Evaluation. A mechanism for quantifying the benefits of safety improvements implemented through a systemic approach

We will spend the majority of time today discussing the first element, but we will also highlight the last two. These discussions will help to meet our second learning outcome “Describe the three elements that make up the Systemic Safety Project Selection Tool”.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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Element 1: Systemic Safety Planning Process

3 Elements:

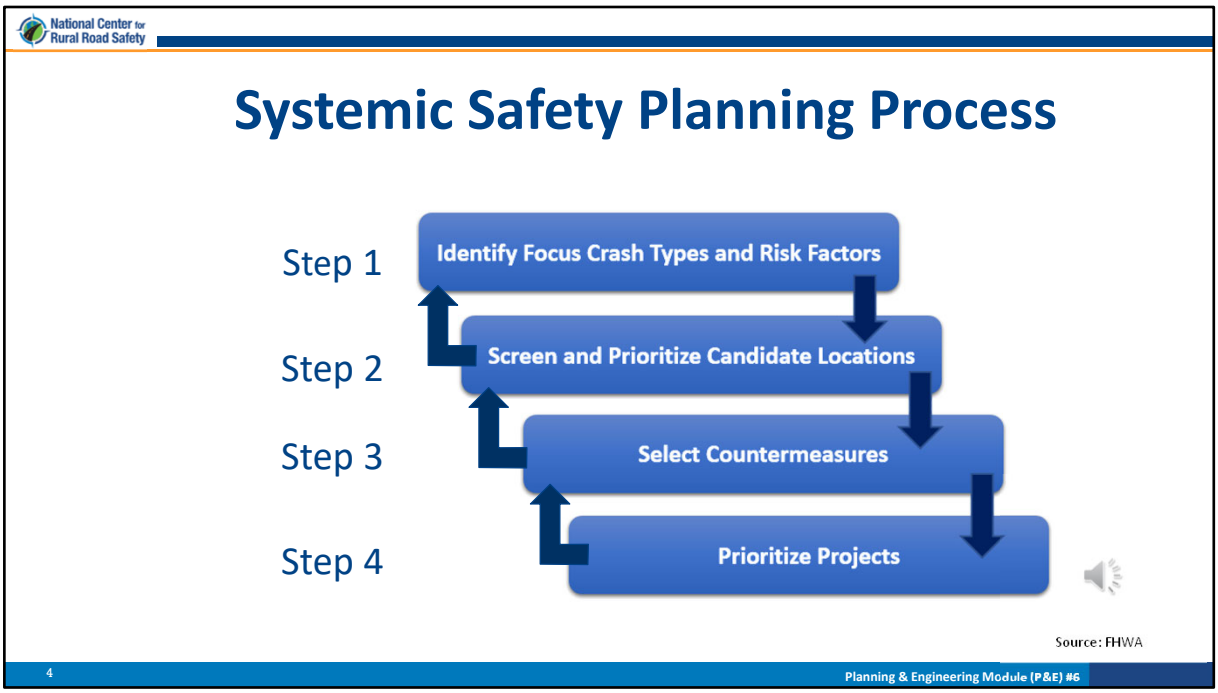
1. Systemic Safety Planning Process
2. A Framework for Balancing Systemic and Traditional Safety Investments
3. Evaluation of a Systemic Safety Program

Source: FHWA

Time on Slide: 1 minute

Suggested Comments/Key Message: So how does one identify those locations with potential for safety improvement? Since there is often not enough funding for low-cost safety improvements across an entire system, you have to prioritize locations with the greatest risk for crashes in the future. Element 1 of the Tool, The Systemic Safety Planning Process helps you with this.

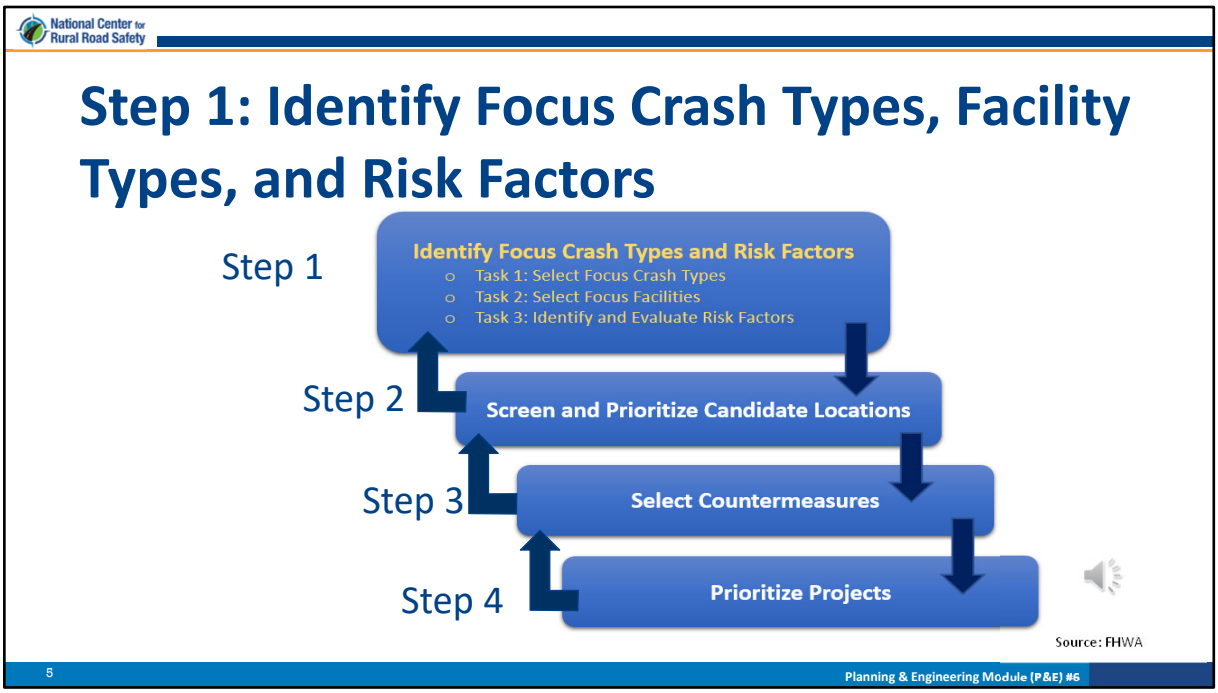
Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf



Suggested Comments/Key Message: The systemic safety planning process is a 4-step process. The process can be scaled based on the availability of technical resources and the quality or quantity of data to support different analytical approaches. The approach is intended to be flexible and adaptable to various agencies to account for differences in data availability, agency practices and regulations, and available resources.

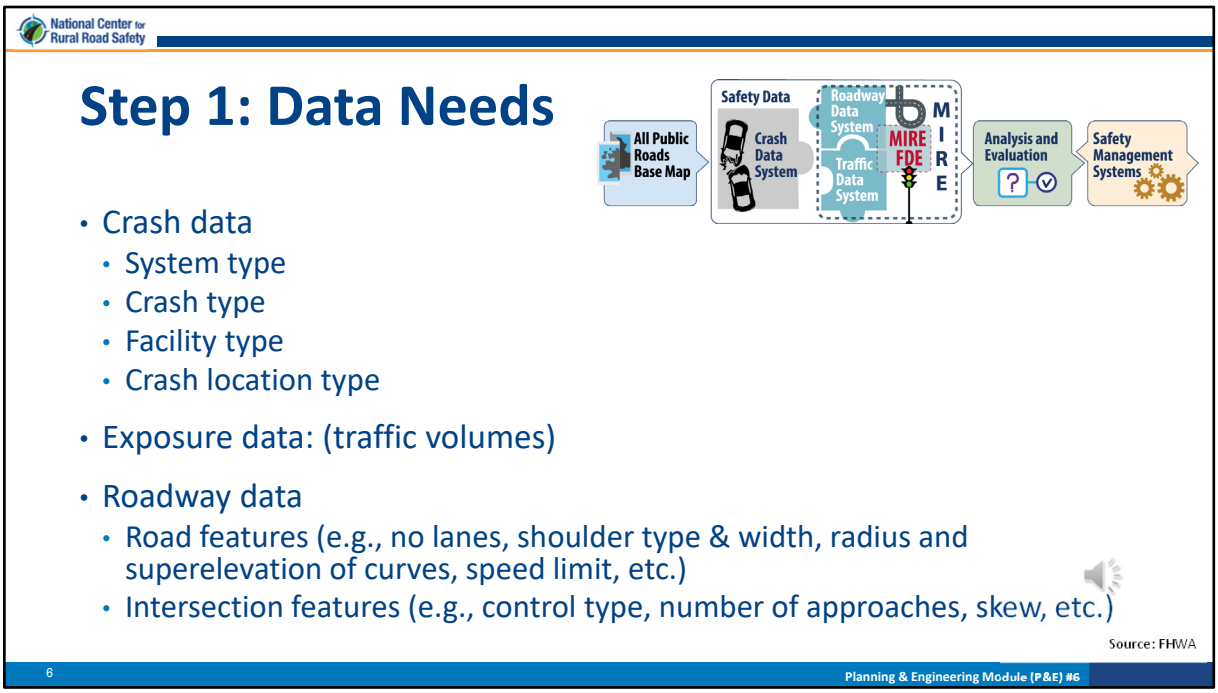
The four steps in the systemic planning process include identifying focus crash types and risk factors, screening and prioritizing candidate locations, selecting countermeasures and prioritizing projects. We will discuss each of these steps in detail on the coming slides to help us meet our third learning outcome "Describe and apply the 4-step systemic planning process".

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf



Suggested Comments/Key Message: As mentioned on the previous slide, the first step in the systemic safety planning process is to identify risk factors that are commonly associated with selected focus crash types that are experienced across a system. In this step you will review system wide (i.e., macro level) crash data to select focus crash types, then in task 2 select focus facilities and then finish the step with task 3 where you will identify factors that are commonly present at the site of these focus crash types which increase the risk of those crashes occurring.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf



Suggested Comments/Key Message: There is a variety of data recommended to have in order to complete step 1 of the systemic safety planning process. It should be noted however, that while this data is recommended, it is not needed to accomplish this step.

Crash data is one of those data elements. It can be utilized to complete all three tasks in step 1. It is recommended that the crash data include at a minimum the:

- System type (e.g. state, local),
- Crash type (e.g. road departure, right angle),
- Facility type (e.g., freeway or collector), and
- Crash location type (e.g., urban vs. rural, intersection vs, segment, curve vs. tangent)

In order to complete risk factor identification and evaluation as well as potentially selecting the focus facilities, one will also need exposure data like traffic volumes and roadway data which are broken into road features and intersection features.

A few of the roadway elements which can be collected include number of lanes, shoulder type and width, radius and superelevation of curves, and speed limit. Intersection features which can be collected may include control type, number of approaches, and skew.

If one is having trouble identifying what data to include, they can begin by referring to the FHWA Model Inventory of Elements (MIRE) Fundamental Data Elements for a list of data elements. One wants to make sure they include any features they think may be associated with the focus crash type.

It should be noted that the collection of data does not have to be completed before starting step 1 and can be done iteratively through the process as the need for additional data elements is identified.

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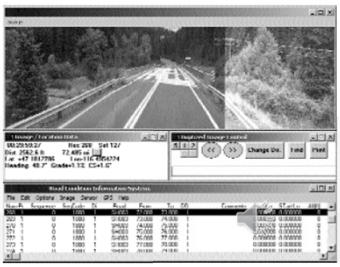
Step 1: Potential Data Sources

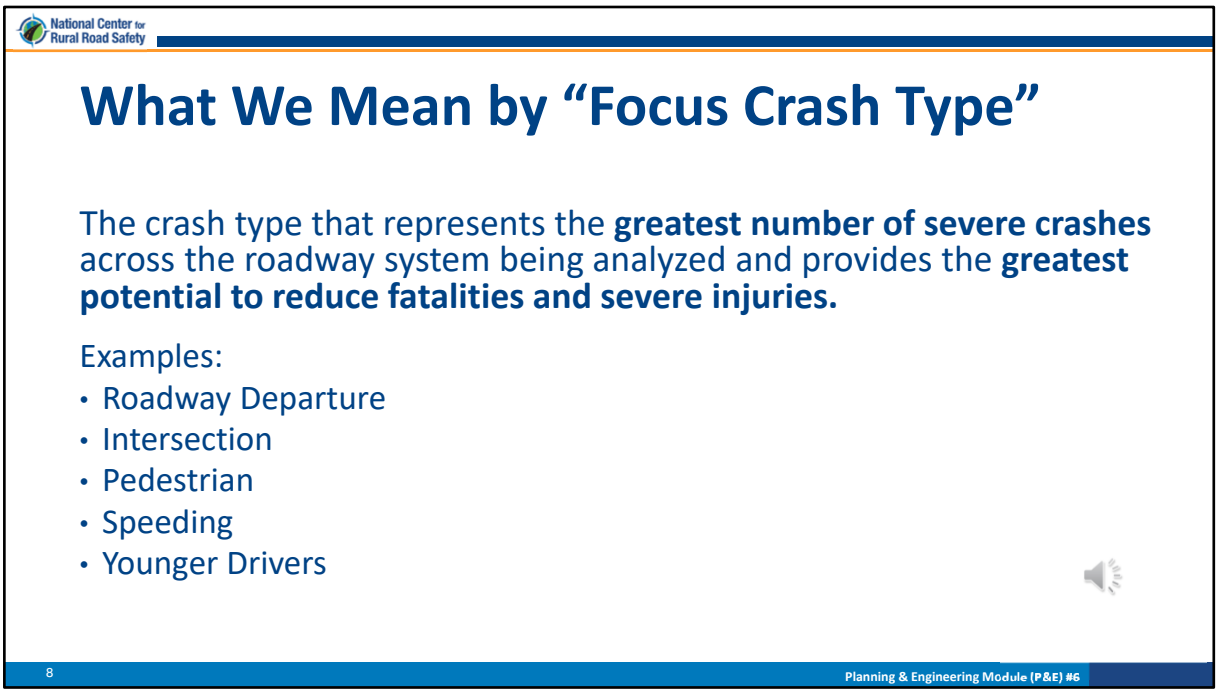
- Crash data
  - Law enforcement
  - State or local database
  - FARS
- Roadway data
  - State or local database
  - Video logs
  - Online aerial imagery
  - Windshield surveys
- Exposure data (AADT)
  - State or local database
  - Traffic counts



Source: FHWA

Suggested Comments/Key Message: Here are a list of potential sources for each of these data types. For crash data **(CLICK)** some examples of sources include law enforcement, state or local database and the Fatality Analysis Reporting System (FARS). For roadway data **(CLICK)** some examples of sources include state or local database, video logs, online aerial imagery, and windshield surveys. For exposure data or Average Annual Daily Traffic (AADT) **(CLICK)** some example sources include state or local database and traffic counts.

Take a moment and drop in the chat pod other sources of data you are aware of and/or use.



Suggested Comments/Key Message: As mentioned previously, the first task in step 1 is to identify the focus crash type, but what do we mean by focus crash types?

A focus crash type is the crash type that represents the greatest number of severe crashes across the roadway system being analyzed and provides the greatest potential to reduce fatalities and injuries. That is really a fancy way of saying which crash types are we seeing most frequently associated with severe or fatal crashes on our roadways. If we are relying on crash data, agencies will likely focus on KA crashes, with K being fatal injury and A being suspected serious injury.

Can you think of potential focus crash types? Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

(CLICK) Some examples include roadway departure, intersection, pedestrian, speeding, and younger drivers.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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How to Identify Focus Crash Type

- Systemwide network analysis
- Strategic safety plans
  - State's Strategic Highway Safety Plan (SHSP) emphasis areas
  - Local Road Safety Plan (LRSP) emphasis areas
  - Safety implementation plans
- Regional and jurisdictional analyses

Source: FHWA

Suggested Comments/Key Message: So how does one go about identifying their focus crash type. One way this can be done is by completing a system wide network analysis where one looks at the crash types that occur on their system and finds those that are over-represented. However, if you do not have crash data available, one could also look at emphasis areas identified in their Strategic Highway Safety Plan, or their Local Road Safety Plan. All of these emphasis areas have been identified through this system-wide network analysis. Some states also have developed safety implementation plans for specific crash types which are overrepresented in their state, so one could look at those as well. You may also want to do a regional or jurisdictional analysis to confirm if your trends are similar to statewide data.

Additionally, we can apply Safe System Approach concepts and look at those crashes that involve significant kinetic energy and therefore result in fatalities and serious injuries such as higher angles, higher speeds, and greater differences in mass.

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Identify Focus Crash Types Example

Fatal and Severe Injury Crashes (2007-2011) Percent by Jurisdiction								
Emphasis Area	Statewide 114,592 mi		State 15,486 mi		County 19,938 mi		City, Town, Village 76,735 mi	
Total Fatal/Serious Injury	100%	63,443	31%	19,819	10%	6,572	45%	28,597
Pedestrian	19%	11,786	9%	1,860	6%	421	28%	8,122
Bicycle	5%	3,390	3%	518	3%	187	8%	2,414
Heavy Vehicle	5%	3,123	6%	1,266	4%	234	4%	1,051
Road Departure	26%	16,668	30%	5,985	44%	2,892	18%	5,128
Intersection	41%	25,791	25%	5,033	30%	1,957	64%	18,270
Head-on and Sideswipe	5%	3,071	7%	1,439	7%	490	3%	887

Source: New York DOT via FHWA

Suggested Comments/Key Message Let's utilize an example to see not only how one can complete a system wide analysis, but also see why sometimes a regional or a jurisdictional analysis is needed. Here is an example which lists the percentage and number of crashes that occur system-wide across the state as well as then broken down by jurisdiction. So, if we were going to identify statewide (2nd column) which crash type to focus on, we would select intersection crashes **(CLICK)** because they represent the highest percentage of fatal and severe injury crashes.

Which crash type would you focus on for state roads (4th column)? **(CLICK)** You would choose road departure crashes because they represent the highest percentage of fatal and severe injury crashes on State Maintained roads.

If we were looking at county roads (6th column), we would choose **(CLICK)** road departure crashes and for city-maintained roads (8th column), we would focus on **(CLICK)** intersection crashes.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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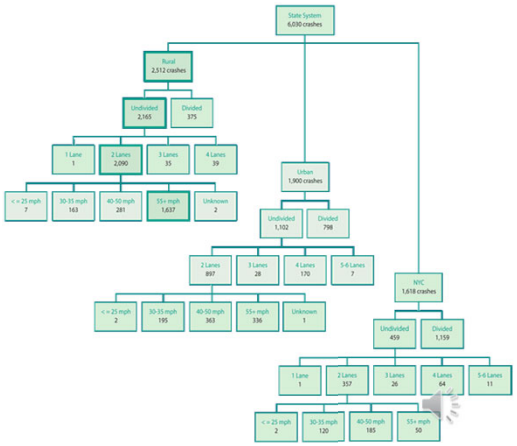
Method for Selecting Focus Facility

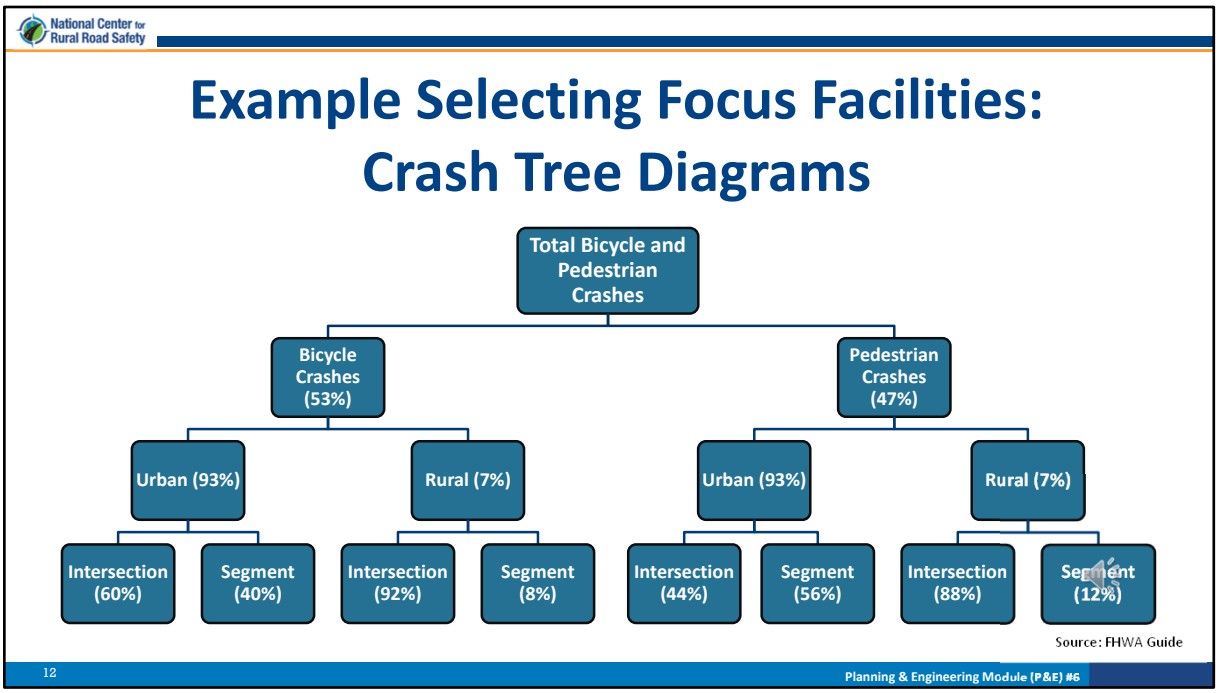
- Crash tree diagrams are recommended method
- Means of breaking down data include:
  - Urban vs. Rural
  - Ownership (state vs. local)
  - On tangent vs. on curve



Source: New York DOT via FHWA

Suggested Comments/Key Message: We use "facility type" to capture both common ways to think about facility type (e.g., rural, two-lane; urban two-lane, rural multilane undivided) as well as site types (e.g., unsignalized intersection, horizontal curve). To determine your focus facility, you could run your data through a variety of pivot tables or data summaries, however the recommended method is to utilize a crash tree. A crash tree is used to break crash data down by various factors. These can include breaking data down between things such as urban vs. rural, state vs. local, or tangent vs. curve. When developing your crash tree, it is recommended you use combinations that will help break the data down quickly. It is also recommended you use binary combinations, so your tree does not get too wide too fast.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf



Suggested Comments/Key Message: Now let's go through an example of how one might utilize a crash tree to select a focus facility. Here we are trying to find which facility to focus on related to bike or pedestrian crashes. We can only look at one facility related to one crash type, so as we go down through each of the levels to determine our focus facility. **(CLICK)** First we choose between bicycle crashes and pedestrian crashes**. (CLICK)** Bicycle crashes have a higher percentage. Next, we choose between urban and rural. **(CLICK)** Urban has the higher percentage at 93%. Lastly, we choose between intersection and segment. **(CLICK)** Intersection has the highest percentage. So, urban intersections would be our focus crash type for our bicycle crashes.

Facilitation Guidance/Interactivity: Present key message using the crash tree on the slide to demonstrate how data is broken down to determine the focus facility. Click four times to animate and as you go through the tree will darken through four animations.

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What we mean by "risk factor"

A qualitative or quantitative measure of the likelihood of a severe crash relative to typical conditions.

Examples:

- Volume
- Alignment
- Intersection Control
- Presence of Shoulders

Source: Walt Catlett, RRSC

Suggested Comments/Key Message: Now that we have determined our focus facility and crash type, we now need to identify and evaluate risk factors as part of task 3. So, what do we mean by risk factors. Risk factors are a qualitative or quantitative measure of the likelihood of a severe crash relative to typical conditions. Risk factors could include: **(CLICK)** traffic volume, roadway alignment, intersection control or the presence of shoulders. Identifying a risk factor does not necessarily mean the observed characteristic "caused" the crashes, as at this point causation may not be determined, instead, risk factors are common characteristics in the roadways where crashes have occurred.

Please take a moment and add risk factors you look for in your jurisdiction. Place them in the chat pod.

Facilitation Guidance/Interactivity: Present first portion of key message and then have the option to ask participant to think of examples and debrief wit the second portion of the key message or just present the second portion of the key message. The list of example will appear through animation when clicked.

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Step 1 - Task 3: Identify and Evaluate Risk Factors

- Identify potential risk factors
  - Roadway and intersection features
  - Traffic volume
  - Transit stops, land use, etc.
- Evaluate risk factors
- Select final risk factors

Photo Source: FHWA

Suggested Comments/Key Message: The first decision to be made for this task is determining which characteristics will be evaluated—in other words, the potential risk factors for the systemic network screening. The initial characteristics are further evaluated before selecting the final characteristics that represent the risk factors. We will discuss potential risk factors associated with common focus crash types and facilities on the upcoming slide.

Selecting characteristics for evaluation might be based on several considerations. Some of these factors include the availability of the data element in existing databases, the ability to quickly gather data if not already available, and the applicability to the focus crash type and facility type (for example, intersection characteristics do not need to be investigated for a systemic lane departure program). *These* will influence the selection of characteristics to review as potential risk factors.

For data elements in an existing database, concerns about quality or the amount of time since the last update also may influence the decision of whether to evaluate. This can be a concern because data that do not reflect current conditions could lead to implementing a countermeasure that is no longer appropriate or the best choice for addressing the focus crash type.

These risk factors are then evaluated to determine if they are overrepresented. Finally, a set of risk factors will be selected based off the results of the evaluation.

Let me give you an example to help you better visualize the result of this process: For example, data analysis could focus on roadway departure crashes on rural two-lane segments and show these crashes were overrepresented on roads with poor road edges, within a specific range of volumes, and a curvilinear alignment.

Identifying risk factors is not an easy task, and there are no right or wrong answers. Safety planners are encouraged to consult national sources, analyze their data, and use their professional judgment to select risk factors that will help distinguish one curve from another, one segment from another, and one intersection from another.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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Potential Risk Factors – Roadway Departure

- Number of lanes
- Lane width
- Shoulder width / type
- Median width / type
- Horizontal curvature
  - Superelevation
  - Delineation
  - Advance warning
  - Speed differential
  - Visual trap
- Pavement condition / friction
- Roadside features
  - Sideslope design
  - Clear zone
- Driveway density
- Other features
  - Rumble strips
  - Lighting
  - On-street parking

Photo Source: Institute for Transportation

Suggested Comments/Key Message: There are many possible risk factors associated with roadway departure crashes. Let's see if you can identify some first, take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

(CLICK) Some examples of the potential risk factors include: number of lanes, lane width, shoulder width and type, median width and type. For horizontal curvature, some potential risk factors include: superelevation, delineation, advance warning, speed differential, and visual trap.

(CLICK) Additional examples include: pavement condition and friction, roadside features like sideslope design and clear zone, driveway density and other features such as rumble strips, lighting, and on-street parking.

This list is not exhaustive but presents some of the common risk factors identified in the Highway Safety Manual and other research.

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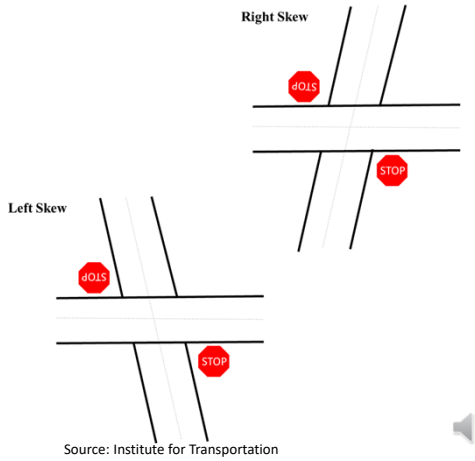
Potential Risk Factors - Intersections

- Traffic control device
- Left-turn or right-turn lanes
- Skew angle
- Advance warning signs
- Located in or near horizontal curve
- Type of development (e.g., commercial)
- Signals
  - Left-turn phasing
  - Number of signal heads vs. number of lanes
  - Backplates
  - Right-turn-on-red
  - Overhead versus pedestal mounted



Source: Institute for Transportation

Suggested Comments/Key Message: There are many possible risk factors associated with intersection crashes. Let's see what you can identify first. What are some risk factors that you look for in intersection crashes? Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

(CLICK) Potential risk factors at intersections include: traffic control devices, left-turn or right-turn lanes, skew angle, advance warning signs, located in or near horizontal curve, type of development (e.g., commercial), and signal details such as left-turn phasing, number of signal heads vs. number of lanes, backplates, right-turn-on-red, and overhead versus pedestal mounted.

Facilitation Guidance/Interactivity: Present key message then ask question below and debrief, bringing up the list and photos of skew by clicking on the slide. Using the slide making note that these are some examples of the potential risk factors that might be present at these locations. This list is not exhaustive but presents some of the common risk factors identified in the Highway Safety Manual and other research.

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Potential Risk Factors - Pedestrians

- Pedestrian volumes
- Controlled or uncontrolled
  - Type (if controlled)
- Marked or unmarked
- Pedestrian signal/type
- Sidewalk presence
- Adjacent land uses
- Lighting

Photo Source: FHWA

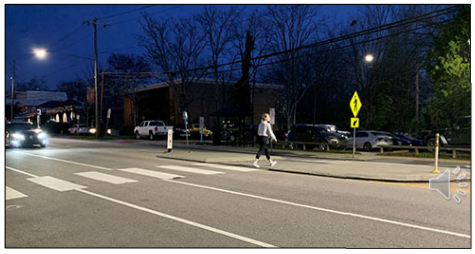

Photo Source: FHWA

Suggested Comments/Key Message: Before we discuss possible risk factors associated with pedestrian crashes. Let's see what you can identify first. Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

(CLICK) Possible pedestrian crash risk factors include: pedestrian volumes, controlled or uncontrolled and the type if controlled, marked or unmarked, pedestrian signal and type, sidewalk presence, adjacent land uses, and lighting.

Facilitation Guidance/Interactivity: Present key message then ask question below and debrief, bringing up the list and photos by clicking on the slide. Using the slide making note that these are some examples of the potential risk factors that might be present at these locations. This list is not exhaustive but presents some of the common risk factors identified in the Highway Safety Manual and other research.

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Potential Risk Factors - Bicyclists

- Bike volumes
- Number of lanes
- Traffic volumes
- Parking type
- Shoulder width
- Bike lane presence
- Access density
- Adjacent land use
- Speed limit

Photo Source: FHWA

Suggested Comments/Key Message: There are many possible risk factors associated with bike crashes. Let's see what you can identify. Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

(CLICK) Possible risk factors associated with bike crashes include: bike volumes, number of lanes, traffic volumes, parking type, shoulder width, bike lane presence, access density, adjacent land use, and speed limit.

Facilitation Guidance/Interactivity: Present key message then ask question below and debrief, bringing up the list and photo on the slide. Using the slide making note that these are some examples of the potential risk factors that might be present at these locations. This list is not exhaustive but presents some of the common risk factors identified in the Highway Safety Manual and other research.

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Other Potential Risk Factors

- Adjacent land use type
  - Schools
  - commercial, or alcohol establishments
- Bus stops (presence and location)
- Railroad crossing
- Others

Photo source: FHWA

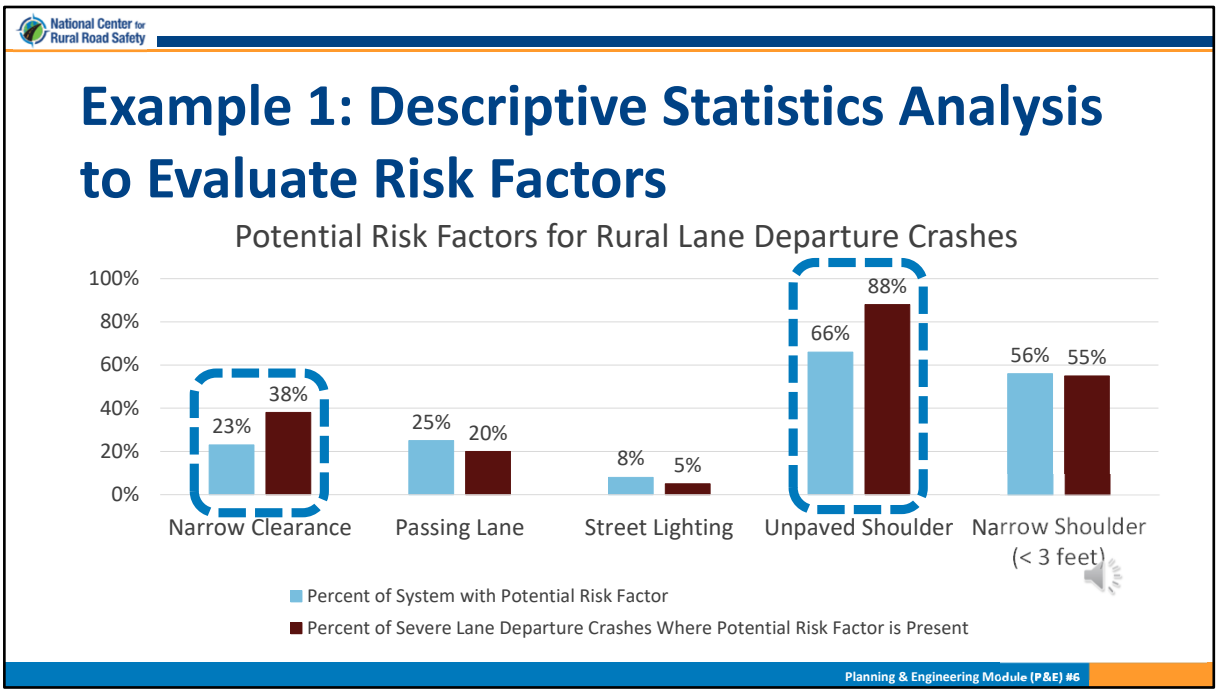
Suggested Comments/Key Message: There are many possible risk factors associated with other general features. These may include: adjacent land use type (e.g., schools, commercial, or alcohol-sales establishments), location and presence of bus stops, presence of nearby railroad crossing or even presence of automated enforcement

Suggested Comments/Key Message: Once you have your list of potential risk factors, you then need to evaluate them to ensure they are actually increasing the risk of our focus crash type on our focus facility type. If you don't have crash and roadway data available to evaluate this, you can look to published literature. This could include the NCHRP 500 series or the CMF Clearinghouse.

However, if you have the data available, you can develop descriptive statistics to determine if the crashes are overrepresented for each potential risk factor, and if so, for what values of that roadway factor. We will go through examples of these on the next few slides.

Facilitation Guidance/Interactivity: To support the use of published literature, you could bring up the example that doctors when evaluating you for risk factors aren't running their own studies each time, they are looking to the published literature to determine those risk factors which have been linked with an increased risk of that specific disease.

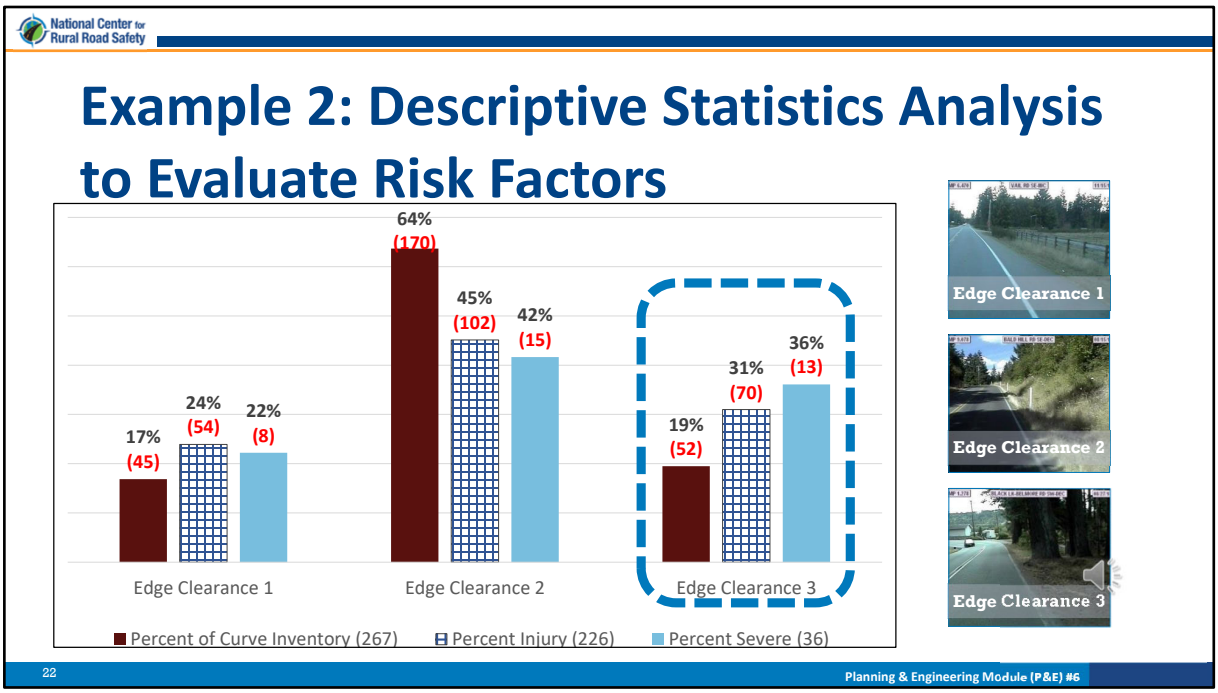
Reference Materials:
CMF Clearinghouse: http://www.cmfclearinghouse.org/
NCHRP 500 Series: http://www.trb.org/Main/Blurbs/152868.aspx



Suggested Comments/Key Message: You can utilize these charts to determine where the percentage of the focus crash type when the risk factor is present is overrepresented compared to the percent of the system which had the potential risk factors present. So, from this you could narrow down your potential risk factors.

From this chart we see that narrow clearance **CLICK** and unpaved shoulder **CLICK** are overrepresented, and therefore they should be included in our final list of risk factors. The other three factors, presence of a passing lane, street lighting and narrow shoulder are all slightly underrepresented and should therefore not be included in the final list of risk factors.

Facilitation Guidance/Interactivity: Use a pointer or draw feature on presentation platform to show which are overrepresented and re-iterate we know they are overrepresented as a higher percentage of lane departure crashes are occurring on these roads than the percentage of these roads that have that risk factor present.



Suggested Comments/Key Message: In addition to just identifying risk factors, if you have the data available, you can further break down potential risk factors and determine which values of a roadway characteristic are associated with an increased risk. In this example, we see that edge clearance has been broken down into three categories ~~of edge clearance~~. On the chart the first dark red bar represents the percentage of curves within the curve inventory which have the specified type of edge clearance. The blue plaid and solid light blue bars then represent the percentage of injury and severe crashes respectively, which have occurred on curves with these edge clearances.

Of the three edge clearances, which one is associated with a greater risk of severe and injury crashes on curves? Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

(CLICK) Edge clearance 3 as the percentage of injury and severe crashes that occur on roads with this edge clearance is quite a bit greater than the percentage of curves with that edge clearance.

Select Final Risk Factors

- Relationship to future crash potential
- Relationship to focus crash and facility types

- Number of risk factors selected?
  - No set rule of thumb
  - Minimum of 2-3 to allow for differentiation between sites
  - 7-10 will take more time, but increases accuracy of rankings

Suggested Comments/Key Message: We have now identified all potential risk factors, evaluated those into a smaller list and then from that list of risk factors we want to select the ones to move forward with to help prioritize our sites. Again, the risk factors selected should relate to the potential for crashes as well as our focus crash and facility types.

(CLICK) You may be asking, how many risk factors do I need to include in my final list. **(CLICK)** There is no set rule of thumb on the exact number. It is recommended that at least 2-3 are utilized to allow for a differentiation in rankings between sites to help prioritize our sites in the next step. Some agencies use 7-10. This will take more time to complete the screening, however it will increase the accuracy of prioritizing sites with the likelihood of future crashes.

This final list of risk factors will then be used as we move onto step 2 where we will screen and prioritize the candidate locations. Before we move onto that, we want to ensure you understand the process for step 1. We are now going to do a quick exercise where we apply what we just discussed.

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Step 1 Exercise

Your county wants to conduct a network level systemic analysis. Utilize the documents in your participant workbook and what we have learned about step 1 in the systemic safety planning process to determine:

1. The focus crash type for county roads
2. The focus facility on your county roads
3. Which risk factors to include in final list using descriptive statistics

Suggested Comments/Key Message: For this exercise imagine you are a county who wants to conduct a network level systemic analysis. You will utilize the information presented on the following slides, along with the information just presented on step 1 to complete this exercise.

We are now going to go through this exercise step-by-step and have you determine the focus crash type for county roads as well as the focus facility for your county roads using a crash tree. We will then use descriptive statistics to determine which risk factors to include in our final list. Let's start.

Facilitation Guidance/Interactivity: This slide describes the activity that will be completed on the next three slides. The instructor can choose to break participants up into groups to complete this exercise or have them work on their own. The instructor can also choose whether to have participants work through the three steps of the activity and then debrief or complete each step and debrief after each step. If completing all steps and then debriefing give them 3-5 minutes and then reconvene and debrief. If going step-by-step, give them 1-2 minutes per step and then debrief. Answers can be given through polls or a chat pod (if virtually) or verbally.

This activity will allow participants to apply a portion of the systemic planning process which helps to meet our third learning outcome “Describe and apply the 4-step systemic planning process”.

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Exercise: Select Focus Crash Type

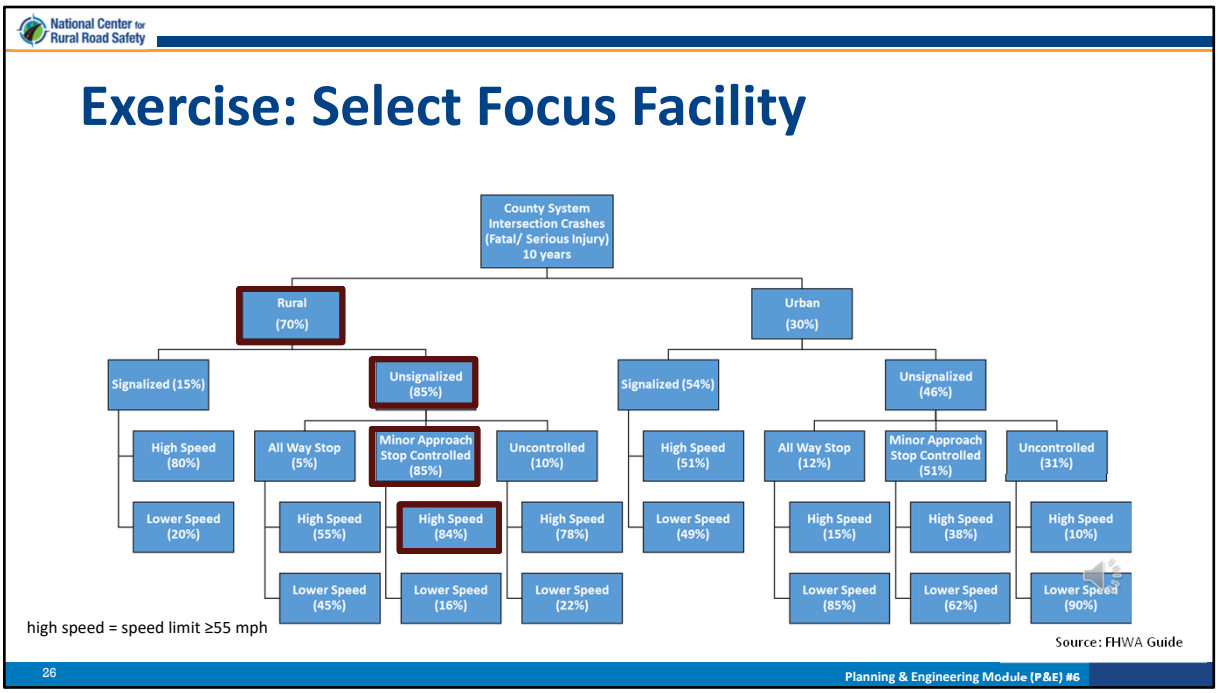
Fatal and Severe Injury Crashes (2011-2020) Percent by Jurisdiction				
Crash Types	Statewide 11,286 mi		County 4,786 mi	
Total Fatal/Serious Injury	100%	512,760	17%	87,170
Pedestrian/Bicycle	4.7%	24,000	8.1%	7,070
Heavy Vehicles	5.3%	27,180	4.2%	3,660
Drug and Alcohol Related	6.1%	31,280	6.9%	6,010
Road Departure	47.5%	243,500	28.5%	24,840
Intersection	28.6%	146,800	42.7%	37,220
Head-on and Sideswipe	7.8%	40,000	9.6%	8,370

Source: FHWA Guide

Suggested Comments/Key Message: In this portion of the exercise, we will select our focus crash type. Which focus crash type should we focus on based off the past 10 years of crash data for our analysis? Remember we are looking at it from a County perspective. Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

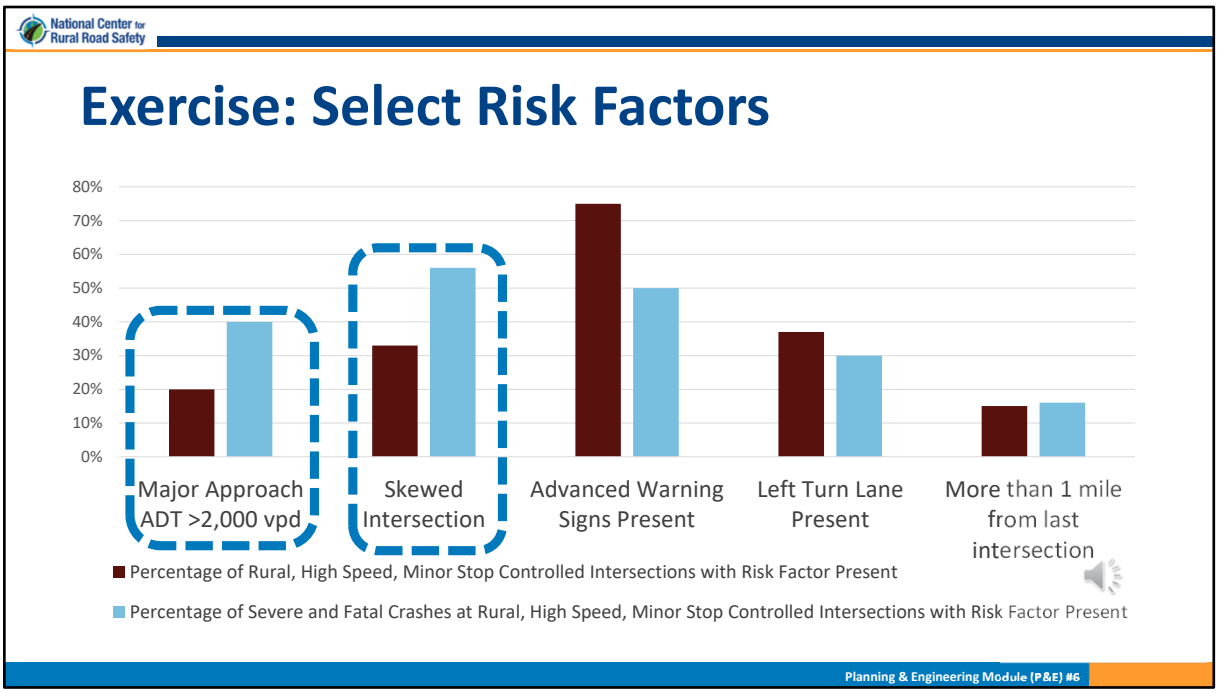
(CLICK) Since we are a county conducting this analysis, we would choose intersection crashes as they are the largest percentage of crashes.

Facilitation Guidance/Interactivity: Present key message, ask question and then click to animate slide and highlight the answer.



Suggested Comments/Key Message: Now we know our focus crash type is intersection crashes. So, we are going to use our crash tree to narrow this down and select a focus facility. What do you think should be our focus facility for our systemic analysis? Take a moment and share your thoughts in the chat pod…. *(Review answers placed in chat pod)*

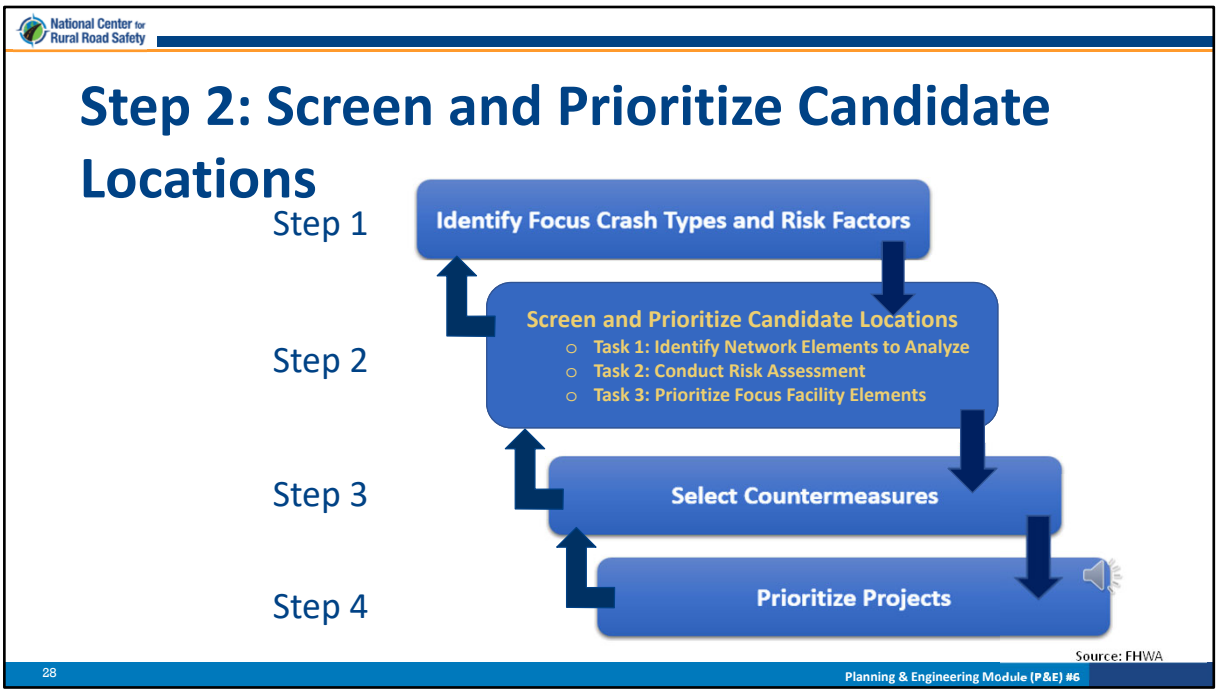
Going through the crash tree, it would be **(CLICK)** Rural, **(CLICK)** unsignalized, **(CLICK)** minor approach stop controlled intersections, and **(CLICK)** high speed.



Time on Slide: 2 -3 minutes

Suggested Comments/Key Message: We have developed a list of potential risk factors for intersection crashes at high speed, rural, minor stop-controlled intersections that we need to narrow down. We have data for five potential risk factors that we need to narrow down to a final list to include only those ~~with~~ that are associated with an increased risk of our focus crash type and facility.

***Ask**:* Which of these factors would you include in your final list of risk factors associated with our focus crash type (intersection crashes) and facility (rural, high speed, two way stop controlled intersections). **Drop your answer in the chat pod**. (if we have time)
***Answer*: CLICK** Major Approach ADT >2,000 vpd and **CLICK** Skewed intersection. Could potentially consider more than 1 mile from last intersection as it is overrepresented, but just slightly. One could potentially see if they can pull out exact distances to see if for instance 2 miles would have a larger percentage of crashes compared to intersections. One should not include advanced warning sign present nor left turn lanes as they are overrepresented.



Suggested Comments/Key Message: In Step 1 we identified a focus crash type and facility and then used those to determine factors associated with an increased risk in these crashes occurring. Now in Step 2 we will utilize the list of risk factors to help develop a prioritized list of potential locations on the roadway system (e.g., segments, curves, intersections) that could benefit from systemic safety improvement projects.

The process to screen and prioritize candidate locations evaluates specific sites (i.e., individual segments, curves, and intersections) at the micro-level. This step is also broken into 3 tasks where one will first identify the appropriate network elements to analyze, then conduct a risk assessment based on the presence of the risk factors identified in step 1 and finally use that to prioritize the elements for potential safety improvements.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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Step 2 – Task 1: Identify Network Elements

- Spot-based (curve or intersections)
- Segments

Photo Source: FHWA

Suggested Comments/Key Message: What do we want to start with, we will want to go through and identify a list of all of the roadways that meet our focus facility type identified in step one. So, for example, if rural two-lane curves was our facility type, we would go through our network and develop a list of curves meeting that criteria. If you were looking at segments, you would want to ensure that the segments are broken up into segments that have consistent geometric features and traffic volumes.

Once we have identified these sites, we would develop a table with a brief description of each site.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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Step 2 – Task 2: Conduct Risk Assessment

- Document physical and traffic characteristics
- Conduct evaluation of network elements

Rank	Corridor	ADT Range	Road Departure Density	Access Density	Curve Critical Radius Density	Edge Risk	Totals
1	144.01	★	★	★	★	★	★★★★★
2	40.04	★	★	★	★	★	★★★★★
3	131.01		★	★	★	★	★★★★
4	9.02	★	★	★	★		★★★★
5	5.06	★	★	★	★		★★★★

Suggested Comments/Key Message: Once we have identified the network elements we will be evaluating, we then need to go through and add some details to that list. These details would be related to our selected risk factors. So, for instance in our exercise (we completed before the break) we had major road ADT greater than 2,000 vpd as a risk factor, therefore in this step we would want to make sure we collected the major road ADT as part of identifying network elements.

So now, we conduct our risk assessment. We would use the information collected in the previous task to evaluate our system for the presence of our risk factors. A simple table such as this is a great method for conducting the risk assessment. Each row is a potential site, and each column represents whether a risk factor is present at the site. If it is present one can denote it by placing a 1 in the cell or an asterisk like above. This would be done for each site and each risk factor and then one would sum up the number to determine a risk score.

You may notice during this step that all your sites have a factor present such as access density above. You may need to remove that risk factor, or you could see if you can break it down and determine if for instance if a certain access density is skewing results and then apply a more restrictive criteria for your risk factor.

Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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Qualitative Approach to Risk Factors

- Using risk factors associated with your roadway network is key to a systemic approach
- Examples of qualitative ratings:
  - Good, Fair, Not-So-Good (curve radius, roadside, etc.)
  - Number per segment, mile, roadway (curves, driveways, intersections, etc.)
  - High, Medium, Low (traffic volumes, crash frequency, etc.)

Suggested Comments/Key Message: When conducting your risk assessment, it is key that you use risk factors associated with your roadway network. However, you may find that you do not have the data available to collect other roadway features, for instance curve radius, or it will take too long to accomplish, or the data you have is suspect. In these cases, you could instead use qualitative data related to the radius, such as if the curve was sharp or gentle.

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Step 2 – Task 3: Prioritize Focus Facility Elements

- Total the number of risk factors present
  - Assign equal weights or relative weights
  - Set threshold for high-priority candidates

Source: Walt Catlett, RRSC

Suggested Comments/Key Message: In the previous task, we determined which risk factors were present on our evaluation sites. In this step, a prioritized lists of segments, horizontal curves, and intersections based on the presence of the selected risk factors is generated. This is done with the understanding that the more risk factors present, the greater the potential for the focus crash type and the higher the priority as a candidate site for safety improvements. This can be done by simply assigning equal weights for each risk factor. Risk factors also can be given relative weights. In this case, particular risk factors—if found to have a stronger association with locations where severe focus crashes occurred—can be given larger weights. We will go through an example of each of these on the next slide.

Once you have your ranked list of sites, you need to prioritize them and come up with a final list. Agencies can adjust the thresholds (the number of risk factors present or a weighted value) used to identify high-priority candidates based on the level of funding anticipated for implementation. To set the threshold, agencies should identify more candidate locations than can be implemented in a given year. This is done so you have a backlog of sites to address with safety improvements and won't require you to run a new analysis every year. Also, sites will likely drop out for other reasons such as other planned improvements or other field conditions.
Reference Materials: https://highways.dot.gov/sites/fhwa.dot.gov/files/2022-06/sspst.pdf

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Example: Screen and Prioritize Candidate Locations

Relative weights

Site	Curve Radius (500'-1000')	Curve Radius Weight	Paved Shoulder (width <2')	Paved Shoulder Weight	History of roadway departure crashes	Roadway Departure Crash History Weight	Stars	Rank
Curve A	*	1		0.5	*	2	*** (3)	2
Curve B		1		0.5		2	(0)	5
Curve C	*	1	*	0.5	*	2	*** (3.5)	1
Curve D		1	*	0.5	*	2	** (2.5)	3
Curve E		1	*	0.5		2	* (0.5)	4

Source: FHWA Guide

Suggested Comments/Key Message: Here we will go through two examples of how equal weights and relative weights can be used to determine our rank. **(CLICK)** As you can see, in the first example each of the three risk factors is given equal weight. This results in the following rankings. **(CLICK)**

If we then instead use relative weights **(CLICK)** as seen here where we have paved shoulder is less than 2' less weight and history of roadway departure crashes given additional weight. **(CLICK)** As you can see this did not greatly change the ranks, likely due to the small number of risk factors present but did result in some differentiation in rankings between curves A & D with curve A increasing in rank when using relative weights.

Facilitation Guidance/Interactivity: Go through the first example (equal weights) and then click on the slide to bring up the rankings. Click again to bring up the relative weights example presenting the second half of the key message. Go through it and then click again to bring up the updated ranks. Highlight how we now see a differentiation in the rank between curves A and D.